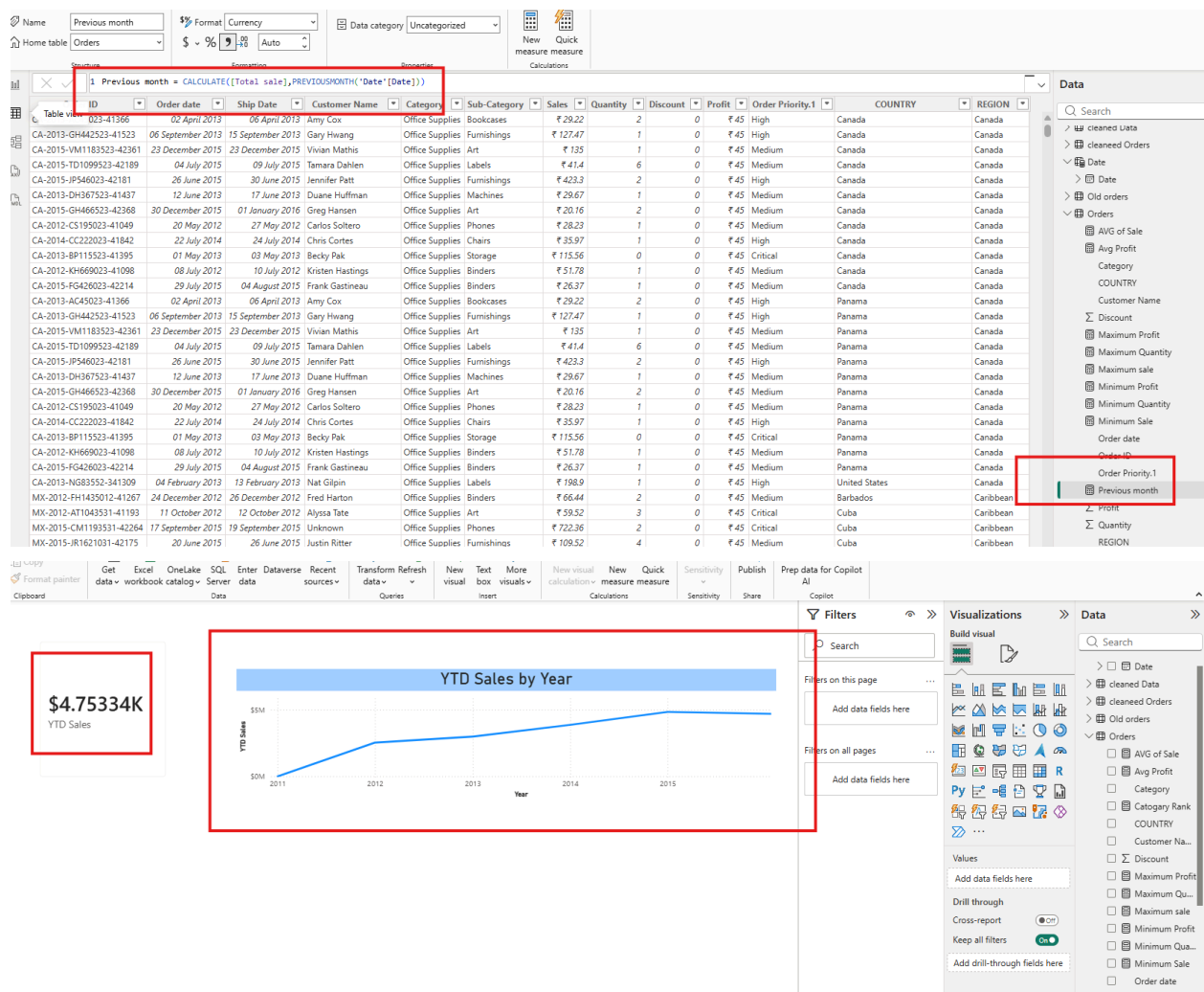


TASK 5 – ADVANCED DAX

- Create measures using **time intelligence functions**.
- Create measures for **comparisons across time periods**.
- Create measures to calculate **growth or percentage change**.
- Create measures to perform **ranking and top N analysis**.
-  **Task:** Develop at least 3–5 advanced measures and apply them in visuals.

Create measures using **time intelligence functions**.

Year-to-Date (YTD) Sales



The screenshot displays the Microsoft Power BI Desktop interface. At the top, the 'Previous month' measure is defined in the DAX formula bar:

```
Previous month = CALCULATE([Total sale],PREVIOUSMONTH('Date'[Date]))
```

The main data table, 'Orders', is visible with columns including ID, Order date, Ship Date, Customer Name, Category, Sub-Category, Sales, Quantity, Discount, Profit, Order Priority, COUNTRY, and REGION. The 'Previous month' measure is highlighted in the 'Data' pane on the right.

Below the data table, a visualization titled 'YTD Sales by Year' is shown. The chart is a line graph with 'Year' on the x-axis (2011 to 2015) and 'YTD Sales' on the y-axis (\$0M to \$5M). The sales show a steady upward trend over the five-year period.

To the left of the chart, a text box displays the total YTD Sales value: **\$4.75334K**.

The 'Visualizations' pane on the right shows the 'Build visual' section with various chart types. The 'Filters' pane on the left shows the 'Filters on this page' and 'Filters on all pages' sections.

The screenshot displays a Power BI interface. On the left, a text box shows the value "₹ 14.2387960..." with the label "Previous year sale" below it. In the center, a bar chart titled "Previous year sale by Order date" shows daily sales data for the years 2013, 2014, and 2015. The y-axis is labeled "Previous year sale" and ranges from ₹ 0K to ₹ 40K. On the right, the Power BI ribbon is visible, showing the "Visualizations" pane with a bar chart icon selected. The "Properties" pane for the selected visual is open, showing the "Title" property set to "Previous year sale by Order date".

Name

Previous year sale

\$ %

Format

Currency

Home table

Orders

\$ %

9.99

Auto

Data category

Uncategorized

New measure

Quick measure

Structure

Formatting

Properties

Calculations

Previous year sale = CALCULATE([Total sale],SAMEPERIODLASTYEAR(Orders[Order date]))

Order date

Ship Date

Customer Name

Category

Sub-Category

Table View

02 April 2013

06 April 2013

Amy Cox

Office Supplies

Bookcases

\$ 29.22

2

0

\$ 45

High

Canada

Canada

15 September 2013

27 May 2012

Gary Hwang

Office Supplies

Furnishings

\$ 127.47

1

0

\$ 45

High

Canada

Canada

23 December 2015

23 December 2015

Vivian Mathis

Office Supplies

Art

\$ 135

1

0

\$ 45

Medium

Canada

Canada

04 July 2015

09 July 2015

Tamara Dahlen

Office Supplies

Labels

\$ 4.14

6

0

\$ 45

Medium

Canada

Canada

26 June 2015

30 June 2015

Jennifer Paff

Office Supplies

Furnishings

\$ 423.3

2

0

\$ 45

High

Canada

Canada

12 June 2013

17 June 2013

Duane Huffman

Office Supplies

Machines

\$ 29.67

1

0

\$ 45

High

Canada

Canada

30 December 2015

01 January 2016

Gary Hansen

Office Supplies

Art

\$ 20.16

2

0

\$ 45

Medium

Canada

Canada

20 May 2012

27 May 2012

Carlos Soltero

Office Supplies

Phones

\$ 28.23

1

0

\$ 45

Medium

Canada

Canada

22 July 2014

24 July 2014

Chris Cortes

Office Supplies

Chairs

\$ 35.97

1

0

\$ 45

High

Canada

Canada

01 May 2013

03 May 2013

Becky Pak

Office Supplies

Storage

\$ 115.56

0

0

\$ 45

Critical

Canada

Canada

08 July 2012

10 July 2012

Kristen Hastings

Office Supplies

Binders

\$ 51.78

1

0

\$ 45

Medium

Canada

Canada

29 July 2015

04 August 2015

Frank Gastineau

Office Supplies

Binders

\$ 26.37

1

0

\$ 45

Medium

Canada

Canada

02 April 2013

06 April 2013

Amy Cox

Office Supplies

Bookcases

\$ 29.22

2

0

\$ 45

High

Panama

Canada

15 September 2013

27 May 2012

Gary Hwang

Office Supplies

Furnishings

\$ 127.47

1

0

\$ 45

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Panama

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Office Supplies

Art

\$ 135

1

0

\$ 45

Medium

Panama

Canada

04 July 2015

09 July 2015

Tamara Dahlen

Office Supplies

Labels

\$ 4.14

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0

\$ 45

Medium

Panama

Canada

26 June 2015

30 June 2015

Jennifer Paff

Office Supplies

Furnishings

\$ 423.3

2

0

\$ 45

High

Panama

Canada

12 June 2013

17 June 2013

Duane Huffman

Office Supplies

Machines

\$ 29.67

1

0

\$ 45

High

Panama

Canada

30 December 2015

01 January 2016

Gary Hansen

Office Supplies

Art

\$ 20.16

2

0

\$ 45

Medium

Panama

Canada

20 May 2012

27 May 2012

Carlos Soltero

Office Supplies

Phones

\$ 28.23

1

0

\$ 45

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Panama

Canada

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Office Supplies

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1

0

\$ 45

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Panama

Canada

01 May 2013

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Office Supplies

Storage

\$ 115.56

0

0

\$ 45

Critical

Panama

Canada

08 July 2012

10 July 2012

Kristen Hastings

Office Supplies

Binders

\$ 51.78

1

0

\$ 45

Medium

Panama

Canada

29 July 2015

04 August 2015

Frank Gastineau

Office Supplies

Binders

\$ 26.37

1

0

\$ 45

Medium

Panama

Canada

04 February 2013

13 February 2013

Nat Gilpin

Office Supplies

Labels

\$ 198.9

1

0

\$ 45

High

United States

Canada

24 December 2012

26 December 2012

Fred Hattin

Office Supplies

Binders

\$ 66.44

2

0

\$ 45

Medium

Barbados

Caribbean

11 October 2012

12 October 2012

Alyssa Tate

Office Supplies

Art

\$ 59.52

3

0

\$ 45

Critical

Cuba

Caribbean

17 September 2015

19 September 2015

Unknown

Office Supplies

Phones

\$ 722.36

2

0

\$ 45

Critical

Cuba

Caribbean

Q Search

cleaned Data

cleaned Orders

Date

Old orders

Orders

AVG of Sale

Avg Profit

Category

COUNTRY

Customer Name

Discount

Maximum Profit

Maximum Quantity

Maximum Sale

Minimum Profit

Minimum Quantity

Minimum Sale

Order date

Order ID

Order Priority

Previous month sale

Previous year sale

Sales Growth:

Name	sales1 growth	Format	Percentage	Data category	Uncategorized	New	Quick						
Home table	Orders	\$	%	9	2								
Structure	Formatting	Properties	Calculations										
1	sales1 growth = DIVIDE([YTD Sales]-[Previous year sale],[Previous year sale])												
Order ID	Order date	Ship Date	Customer Name	Category	Sub-Category	Sales	Quantity	Discount	Profit	Order Priority.1	COUNTRY	REGION	sales
CA-2013-AC45023-41366	02 April 2013	06 April 2013	Amy Cox	Office Supplies	Bookcases	₹ 29.22	2	0	₹ 45	High	Canada	Canada	
CA-2013-GH442523-41523	06 September 2013	15 September 2013	Gary Hwang	Office Supplies	Furnishings	₹ 127.47	1	0	₹ 45	High	Canada	Canada	
CA-2015-TD1099523-42189	04 July 2015	09 July 2015	Tamara Dahlen	Office Supplies	Labels	₹ 41.4	6	0	₹ 45	Medium	Canada	Canada	
CA-2015-JPS46023-42181	26 June 2015	30 June 2015	Jennifer Patt	Office Supplies	Furnishings	₹ 423.3	2	0	₹ 45	High	Canada	Canada	
CA-2013-DH367523-41437	12 June 2013	17 June 2013	Duane Huffman	Office Supplies	Machines	₹ 29.67	1	0	₹ 45	Medium	Canada	Canada	
CA-2015-GH466523-42368	30 December 2015	01 January 2016	Greg Hansen	Office Supplies	Art	₹ 20.16	2	0	₹ 45	Medium	Canada	Canada	
CA-2012-CS195023-41049	20 May 2012	27 May 2012	Carlos Soltero	Office Supplies	Phones	₹ 28.23	1	0	₹ 45	Medium	Canada	Canada	
CA-2014-CC22023-41842	22 July 2014	24 July 2014	Chris Cortes	Office Supplies	Chairs	₹ 35.97	1	0	₹ 45	High	Canada	Canada	
CA-2013-BP115523-41395	01 May 2013	03 May 2013	Becky Pak	Office Supplies	Storage	₹ 115.56	0	0	₹ 45	Critical	Canada	Canada	
CA-2012-KH669023-41098	08 July 2012	10 July 2012	Kristen Hastings	Office Supplies	Binders	₹ 51.78	1	0	₹ 45	Medium	Canada	Canada	
CA-2015-FG426023-42214	29 July 2015	04 August 2015	Frank Gastineau	Office Supplies	Binders	₹ 26.37	1	0	₹ 45	Medium	Canada	Canada	
CA-2013-AC45023-41366	02 April 2013	06 April 2013	Amy Cox	Office Supplies	Bookcases	₹ 29.22	2	0	₹ 45	High	Panama	Canada	
CA-2013-GH442523-41523	06 September 2013	15 September 2013	Gary Hwang	Office Supplies	Furnishings	₹ 127.47	1	0	₹ 45	High	Panama	Canada	
CA-2015-VM1183523-42361	23 December 2015	23 December 2015	Vivian Mathis	Office Supplies	Art	₹ 135	1	0	₹ 45	Medium	Panama	Canada	
CA-2015-TD1099523-42189	04 July 2015	09 July 2015	Tamara Dahlen	Office Supplies	Labels	₹ 41.4	6	0	₹ 45	Medium	Panama	Canada	
CA-2015-JPS46023-42181	26 June 2015	30 June 2015	Jennifer Patt	Office Supplies	Furnishings	₹ 423.3	2	0	₹ 45	High	Panama	Canada	
CA-2013-DH367523-41437	12 June 2013	17 June 2013	Duane Huffman	Office Supplies	Machines	₹ 29.67	1	0	₹ 45	Medium	Panama	Canada	
CA-2015-GH466523-42368	30 December 2015	01 January 2016	Greg Hansen	Office Supplies	Art	₹ 20.16	2	0	₹ 45	Medium	Panama	Canada	
CA-2012-CS195023-41049	20 May 2012	27 May 2012	Carlos Soltero	Office Supplies	Phones	₹ 28.23	1	0	₹ 45	Medium	Panama	Canada	
CA-2014-CC22023-41842	22 July 2014	24 July 2014	Chris Cortes	Office Supplies	Chairs	₹ 35.97	1	0	₹ 45	High	Panama	Canada	
CA-2013-BP115523-41395	01 May 2013	03 May 2013	Becky Pak	Office Supplies	Storage	₹ 115.56	0	0	₹ 45	Critical	Panama	Canada	
CA-2012-KH669023-41098	08 July 2012	10 July 2012	Kristen Hastings	Office Supplies	Binders	₹ 51.78	1	0	₹ 45	Medium	Panama	Canada	
CA-2015-FG426023-42214	29 July 2015	04 August 2015	Frank Gastineau	Office Supplies	Binders	₹ 26.37	1	0	₹ 45	Medium	Panama	Canada	
CA-2013-NG83523-341309	04 February 2013	13 February 2013	Nat Gilpin	Office Supplies	Labels	₹ 198.9	1	0	₹ 45	High	United States	Canada	
MX-2012-FH1435012-41267	24 December 2012	26 December 2012	Fred Harton	Office Supplies	Binders	₹ 66.44	2	0	₹ 45	Medium	Barbados	Caribbean	
MX-2012-AT1043531-41193	11 October 2012	12 October 2012	Alyssa Tate	Office Supplies	Art	₹ 59.52	3	0	₹ 45	Critical	Cuba	Caribbean	
MX-2015-CM1193531-42264	17 September 2015	19 September 2015	Unknown	Office Supplies	Phones	₹ 722.36	2	0	₹ 45	Critical	Cuba	Caribbean	

Category

COUNTRY

CATEGORY RANK

Customer Name

Discount

Maximum Profit

Maximum Quantity

Maximum sale

Minimum Profit

Minimum Quantity

Minimum Sale

Order date

Order ID

Order Priority.1

Previous month sale

Previous year sale

Profit

Quantity

REGION

Sales

sales1 growth

Ship Date

Sub-Category

Rank by Category:

new chat

Pages

Library

Logins

Projects

Index

More

Notes

Cleaning

Used Data

Column

Binding and

Priority

Data Class

Query

Excel Explorer

Dataset

IF Statement

et Analysis

Name Orders

Manage relationships

New measure

Quick measure column

New table

Mark as date table

Structure

Relationships

Calculations

Category Rank = rankx(ALL(Orders[Category]),[Total sale])

Table view

Order date

Ship Date

Customer Name

Category

Sub-Category

Sales

Quantity

Discount

Profit

Order Priority.1

COUNTRY

REGION

sales

Q Search

Order Id

Order Priority.1

Previous month sale

Previous year sale

Profit

Quantity

REGION

Sales

sales with quantity

sales1 growth

Ship Date

Sub-Category

Total no. of Order Id

Total Profit

Total Quantity

Total sale

YTD Sales

return

people

Order Id

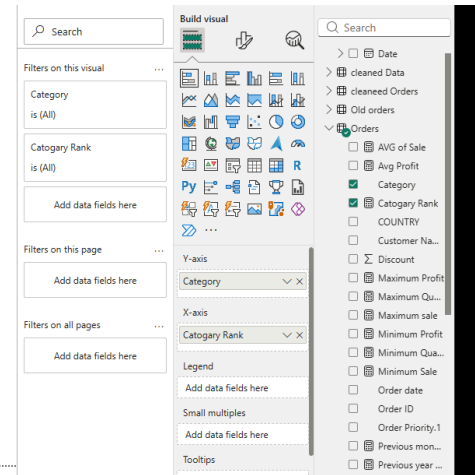
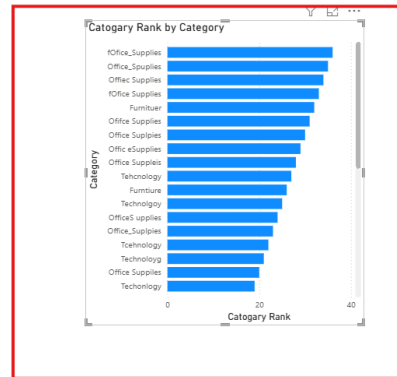
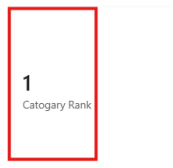
Region

Returned

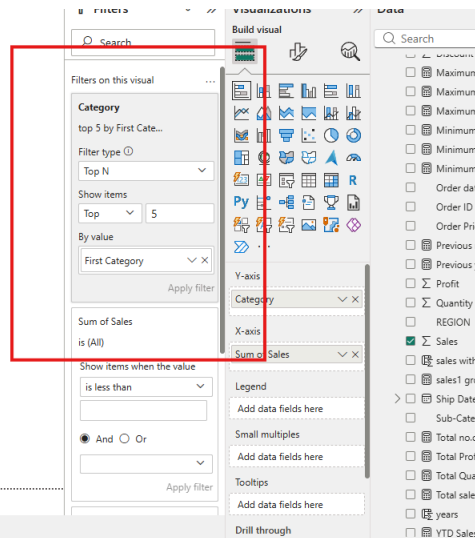
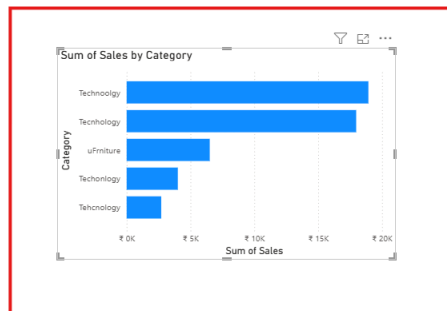
Total no. of order Return

Table (2)

Table 2



Top 5 Analysis



Observations

1. **YTD Sales** shows the total sales from the beginning of the year to the selected date.
2. **Previous Year Sales** helps compare current sales with the same period of the previous year.
3. **Sales Growth %** clearly shows whether sales have increased or decreased.
4. **Category Rank** identifies the highest and lowest performing categories based on sales.
5. **Top 5 Analysis** helps identify the best-selling categories or products quickly.